CSE108 – Computer Programming Laboratory Lab #9

Date: April 24, 2026

In this lab, you will complete the implementation of a **student grading management system**.

The starter code **(Lab9_Code.c)** provides the complete program structure, including **helper functions** and the **main()** driver. **Your task is to implement 5 functions marked with TODO comments.**

NOTE: Only modify the five functions marked with TODO comments in Lab9_Code.c. Do not change any other code in the file.

The TODO comments in each function provide detailed step-by-step instructions. Follow these instructions to complete the implementations of the functions. This document also have the detailed instructions as given below.

TODO 1. ***void get_course_details_from_user(course_t *course);***

1) Ask the user for the course code (it is a string) and store it appropriately using course variable.
2) Ask the user for the number of exams (it is an integer) and store it appropriately using course variable.
3) Loop through all exams in course using exam_count.
4) Inside loop, for each exam, ask the user for the exam name and store it appropriately using course variable.
5) Inside loop, for each exam, ask the user for the exam's weight and store it appropriately using course variable.

NOTE: Inside the loop, just before calling each scanf function, you might need to use this line: *fflush(stdout);* especially, if you are trying to use the scanf just after you use printf in a loop in this function.

TODO 2. ***void get_student_details_from_user(student_t *student, const course_t *courses, int course_count);***

1) Ask the user for the student name and store it appropriately using student variable.
2) Loop through all courses using course_count variable.
3) Loop through all exams of each course.
4) Ask the user for exam grades of the course and store it appropriately using student variable.

TODO 3. ***void list_students(student_t *students, int student_count, const char *course_code, course_t *courses, int course_count, double grade_threshold);***

1) Find course_index for the given course_code. To do this, call index_of_course_by_code function and assign its return value to a variable.
2) Loop through all students using student_count variable.
3) For each student, compute the weighted average of their grades for the course whose index was found above. Call weighted_average_for_course function.
4) Print the student name if the return value of weighted_average_for_course function is greater than grade_threshold.

TODO 4. ***double weighted_average_for_course(const student_t *student, int course_index, const course_t *courses);***

1) Define two double variable named weighted_sum and total_weight.
2) Initialize weighted_sum and total_weight to 0.0.
3) Loop through all exams in courses[course_index] using exam_count.
4) Inside the loop, calculate weighted_sum and accumulate it.
5) Inside the loop, calculate total_weight and accumulate it.
6) Return weighted_sum / total_weight.

NOTE: weighted_sum += grades * weights. Use student parameter to access grades' values. Use courses[course_index] to access weights.

NOTE: total_weight += weights. Use courses[course_index] to access weights.

TODO 5. ***void print_course(const course_t *course);***

1) Print the course code and exam count of the course. Use the course variable given as parameter.
2) Loop through all exams in course using exam_count.
3) In each iteration, print the exam name of the course and its weight.

```
========== Q1: get_course_details_from_user ==========
Enter the number of courses to be defined: 2
Enter course code: CSE102
Enter number of exams: 2
Exam 1 name: midterm
Exam 1 weight: 25
Exam 2 name: final
Exam 2 weight: 40
Enter course code: CSE101
Enter number of exams: 3
Exam 1 name: midterm
Exam 1 weight: 30
Exam 2 name: quiz
Exam 2 weight: 20
Exam 3 name: final
Exam 3 weight: 40

========= print_course ==========
Course CSE102 (2 exam(s))
  [1] midterm              weight: 25.00
  [2] final                weight: 40.00

Course CSE101 (3 exam(s))
  [1] midterm              weight: 30.00
  [2] quiz                 weight: 20.00
  [3] final                weight: 40.00
```

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========== Q2: get_student_details_from_user ==========
Enter number of students: 2
Enter student name: ahmet
Enter grades for course CSE102:
Grade of exam midterm: 50
Grade of exam final: 40
Enter grades for course CSE101:
Grade of exam midterm: 60
Grade of exam quiz: 70
Grade of exam final: 90
Enter student name: burak
Enter grades for course CSE102:
Grade of exam midterm: 45
Grade of exam final: 80
Enter grades for course CSE101:
Grade of exam midterm: 40
Grade of exam quiz: 65
Grade of exam final: 85

========== Q3: list_students (threshold 50.0) ==========
burak
```